Quantum Scientific Computing (QSC2026)

Homework 1

Assigned: Due: November 12, 2026

Part I — Foundations (Lecture 1)

Problem 1. A single qubit: amplitudes, Bloch vector, and measurement. *(10 pts)* Let $|\psi\rangle = \cos\frac{\pi}{8}\,|0\rangle + \mathrm{e}^{\mathrm{i}\pi/4}\sin\frac{\pi}{8}\,|1\rangle$.

(a) Check that $|\psi\rangle$ is normalized and give the probabilities of outcomes 0 and 1 in a computational-basis measurement.

(b) Compute the expectation values $\langle\psi|X|\psi\rangle$, $\langle\psi|Y|\psi\rangle$, $\langle\psi|Z|\psi\rangle$ and hence the Bloch vector; verify it has unit length.

(c) Find the probability of outcome "+" when $|\psi\rangle$ is measured in the $\{|+\rangle, |-\rangle\}$ basis, and relate your answer to $\langle\psi|X|\psi\rangle$.

(d) *Sampling cost.* In practice $\langle Z\rangle$ is not read off exactly but *estimated*: each computational-basis measurement returns the eigenvalue $+1$ (outcome 0) or -1 (outcome 1), and one averages the results of N independent repetitions, $\hat{Z} = \frac{1}{N}\sum_{i=1}^{N} m_i$. The single-shot outcome $m_i \in \{\pm 1\}$ is random with variance $1 - \langle Z\rangle^2$, so $\hat{Z}$ fluctuates about $\langle Z\rangle$ with a standard error that shrinks only as $1/\sqrt{N}$ (the "shot-noise" or standard quantum limit). Using this scaling, estimate how many repetitions N are needed to determine $\langle Z\rangle$ to additive error $\epsilon = 0.05$ at about 95% confidence. State the scaling law (how N depends on ϵ) and show how your number follows from it.

Problem 2. Tensor products, entanglement, and the separability test. *(10 pts)* For each two-qubit state, use the separability criterion of §1.2.2 (a state $a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$ is a product state iff $ad = bc$) to decide whether it is a product state or entangled. If it is a product state, write it as $|u\rangle \otimes |v\rangle$; if entangled, say so.

(a) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$;

(b) $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$;

(c) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$.

(d) Prove directly that the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ admits no factorization $|u\rangle \otimes |v\rangle$.

Problem 3. Gates, circuits, reversibility, and no-cloning. *(10 pts)*

(a) Compute $\mathrm{CNOT}\,(H \otimes \mathbb{I})\,|00\rangle$ and identify which Bell state it is.

(b) Show the Toffoli gate acts as $|a, b, 0\rangle \mapsto |a, b,\, a \wedge b\rangle$, a reversible AND. Why does reversibility require writing the result onto a fresh (ancilla) qubit rather than overwriting an input?

(c) Exhibit a two-qubit unitary U and a blank state $|s\rangle$ with $U|0\rangle|s\rangle = |0\rangle|0\rangle$ and $U|1\rangle|s\rangle = |1\rangle|1\rangle$, and explain why U fails to copy a general unknown qubit $\alpha|0\rangle + \beta|1\rangle$—i.e. why this is consistent with the no-cloning theorem.

Problem 4. Quantum teleportation, traced amplitude by amplitude. *(10 pts)* Alice wishes to teleport $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to Bob; they share $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

(a) Write the three-qubit state before Alice's Bell measurement, expanded over her two-qubit measurement basis, and read off Bob's conditional (pre-correction) state for each of the four outcomes (m_1, m_2).

(b) For the outcome $(m_1, m_2) = (1, 1)$, state the correction gate Bob applies and verify it restores $|\psi\rangle$.

(c) Explain why the two classical bits are indispensable—that is, why Bob's qubit carries no information about $|\psi\rangle$ until the bits arrive (the no-signaling point).

Part II — Query algorithms and Grover search (Lecture 3)

Problem 5. Phase kickback, Deutsch, and Deutsch–Jozsa. *(10 pts)*

(a) For a Boolean f, show the phase-oracle identity $U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$, where $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

(b) List the four functions $f : \{0,1\} \to \{0,1\}$, classify each as constant or balanced, and compute by hand the first-qubit measurement outcome produced by Deutsch's algorithm for each. Confirm it separates constant from balanced in a *single* query.

(c) In Deutsch–Jozsa on n bits, what is the probability of measuring the all-zeros outcome 0^n when f is constant? when f is balanced? State the promise that makes the one-query decision exact.

Problem 6. Grover search: reflections, geometry, and iteration count. *(10 pts)* Take $N = 16$ with a single marked item x_0, uniform start $|\psi_0\rangle = H^{\otimes 4}|0^4\rangle$.

(a) Writing $\sin(\theta/2) = 1/\sqrt{N}$, give θ and explain, using the two reflections $R_{x_0} = \mathbb{I} - 2|x_0\rangle\langle x_0|$ and $R_{\psi_0} = 2|\psi_0\rangle\langle\psi_0| - \mathbb{I}$, why one Grover iterate $G = R_{\psi_0}R_{x_0}$ rotates the state by θ in the plane $\mathrm{span}\{|x_0\rangle, |\psi_0\rangle\}$.

(b) The success probability after k iterates is $p_k = \sin^2((2k+1)\theta/2)$. Evaluate $p_0, \ldots, p_5$, find the optimal k^*, and describe quantitatively what happens at $k = 4, 5$ (over-rotation).

(c) For general $N = 2^n$ with one marked item, give the leading-order optimal iteration count k^* and the resulting query complexity.

Part III — Quantum Fourier transform and phase estimation (Lecture 4)

Problem 7. The Hadamard test. *(10 pts)* Let U be a unitary and $|\psi\rangle$ a state, with the standard Hadamard-test circuit: Hadamard on the ancilla, controlled-U, Hadamard, then measure the ancilla.

(a) Show the ancilla outcome has $p(0) = \frac{1}{2}(1 + \Re\langle\psi|U|\psi\rangle)$, and describe the one-gate modification that instead yields $\Im\langle\psi|U|\psi\rangle$.

(b) Take $U = S = \mathrm{diag}(1, \mathrm{i})$ and $|\psi\rangle = |+\rangle$. Compute $\langle\psi|U|\psi\rangle$ directly, then confirm it is consistent with the $p(0)$ (and the imaginary-part variant) of part (a).

Problem 8. QFT and phase estimation. *(10 pts)*

(a) Write the product (tensor-factor) form of $\mathrm{QFT}\,|j\rangle$ on $n = 3$ qubits and count the Hadamard and controlled-phase gates in the circuit.

(b) Run standard QPE with $t = 3$ ancillas on an eigenstate with phase $\varphi = 0.011_2 = \frac{3}{8}$. Write the ancilla state after the controlled powers, identify it as $\mathrm{QFT}\,|k\rangle$ for the correct k, and give the measurement outcome. Why is the readout exact here?

(c) Using the precision/qubit-count relation of §4.3–4.4, determine the number of ancilla qubits t needed to read $n = 6$ bits of φ with success probability at least 0.9. State the formula, then evaluate it.

Part IV — Applications of phase estimation (Lecture 5)

Problem 9. Ground-state energy estimation. *(10 pts)* Let H have eigenpairs $\{(E_j, |E_j\rangle)\}$ and set $U = \mathrm{e}^{-\mathrm{i}Ht}$ for a fixed $t > 0$ chosen so that every $E_j t/(2\pi) \in [0, 1)$.

(a) Show $|E_j\rangle$ is an eigenstate of U with eigenphase $\varphi_j = -E_j t/(2\pi) \bmod 1$, so that a QPE estimate of φ_j yields an estimate of E_j. Give the formula recovering E_j from φ_j.

(b) One prepares an initial guess $|\phi\rangle = \sum_j c_j\,|E_j\rangle$ with ground-state overlap $p_0 = |c_0|^2$ and runs QPE on U. Explain why a single run returns the ground-state phase only with probability p_0, and compute how many independent runs R guarantee seeing it at least once with probability ≥ 0.99 when $p_0 = 0.4$.

(c) Briefly: why does this energy-estimation scheme *not* require a spectral gap, whereas preparing the ground state does?

Problem 10. Amplitude estimation and the quadratic advantage. *(10 pts)* Let $A\,|0^n\rangle = \sqrt{P}\;|\phi_{\mathrm{good}}\rangle + \sqrt{1-P}\;|\phi_{\mathrm{bad}}\rangle$ and write $P = \sin^2(\theta/2)$.

(a) State why phase estimation of the Grover iterate $G = R_{\phi_0} R_{\mathrm{good}}$ (with $R_{\phi_0} = 2\,|\phi_0\rangle\langle\phi_0| - \mathbb{I}$, $R_{\mathrm{good}} = \mathbb{I} - 2\,|\phi_{\mathrm{good}}\rangle\langle\phi_{\mathrm{good}}|$, $|\phi_0\rangle = A\,|0^n\rangle$) estimates θ, and hence P. What are the eigenvalues of G on the invariant plane?

(b) To estimate P to additive error $\epsilon = 10^{-3}$, compare the number of uses of A (and of G) for amplitude estimation, $\mathcal{O}(1/\epsilon)$, against classical Monte Carlo sampling, $\mathcal{O}(1/\epsilon^2)$. Give both counts.

(c) Combine with Problem 7: amplitude estimation applied to the Hadamard-test success probability $p(0) = \frac{1}{2}(1 + \Re\,\langle\psi|U|\psi\rangle)$ estimates $\Re\,\langle\psi|U|\psi\rangle$. What precision scaling in queries does this achieve, and why is it a quadratic improvement over plain repetition of the Hadamard test?

Problems drawn from Lectures 1 (§1.1–§1.6), 3 (§3.1–§3.3), 4 (§4.1–§4.3), and 5 (§5.1–§5.2) of the Quantum Scientific Computing course notes.